Linear Algebra and Its Applications

Section 8.4: Hyperplanes

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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8.4 Hyperplanes

A linear functional has scalar output: $f : \mathbb{R}^n \to \mathbb{R}$. It can be written $f(x) = n^T x$ for a fixed vector n. When $n \neq 0$, the level set $[f : d] = \{x : n^T x = d\}$ is a hyperplane. Its two open sides have $f(x) < d$ and $f(x) > d$. To separate sets weakly, put them in opposite closed sides; to separate strictly, put them in opposite open sides. Here "open" and "closed" refer to the ambient $\mathbb{R}^n$, not just the set's own line or plane. In Euclidean space compact means closed and bounded.

Practice problem 8.4.P1: An explicit line where two planes intersect

The normal equations are $x_1 + x_2 - 2x_3 = -3$ and $-2x_1 + x_2 + 3x_3 = 7$, using $n_1 \cdot p_1 = -3$ and $n_2 \cdot p_2 = 7$. Subtracting the second equation from the first after elimination gives the reduced equations $x_1 - (5/3)x_3 = -10/3$, $x_2 - (1/3)x_3 = 1/3$. Thus their intersection is

$$x = (-10/3, 1/3, 0) + t(5/3, 1/3, 1), \qquad t \in \mathbb{R}.$$

Substitution verifies both plane equations for every t, and the single free variable shows that every point in the intersection has this form.

Exercise 8.4.1: An implicit equation for a line

The direction from $(-1, 4)$ to $(3, 1)$ is $(4, -3)$. Choose the perpendicular normal $n = (3, 4)$, since $3(4) + 4(-3) = 0$. Then $d = n \cdot (-1, 4) = -3 + 16 = 13$, so take $f(x, y) = 3x + 4y$ and $d = 13$. Both points satisfy this equation, and its nonzero normal makes it the unique line through those two distinct points.

Exercise 8.4.2: A second line functional

The direction between $(1, 4)$ and $(-2, -1)$ is $(-3, -5)$. A perpendicular normal is $(5, -3)$; its dot product with the direction is $-15 + 15 = 0$. At the first point it has value $5 - 12 = -7$. Thus $f(x, y) = 5x - 3y$ and $d = -7$ give the required line $[f : d]$.

Exercise 8.4.3: Open, closed, or neither

(a) $y > 0$ is **open**, not closed. Every point has a ball staying above the axis; $(0, 1/k)$ tends to the excluded origin.

(b) The segment $x = 2$, $1 \leq y \leq 3$ is **closed**, not open. It includes its endpoints and is the intersection of closed constraints. Every disk about a point on it contains points with $x \neq 2$.

(c) The segment $x = 2$, $1 < y < 3$ is **neither**. It has no ambient open disk, and $(2, 1 + 1/k)$ tends to an excluded endpoint.

(d) The positive branch $xy = 1$, $x > 0$ is **closed**, not open. For a convergent sequence on it, the limit satisfies $xy = 1$ by continuity; its x cannot be zero or negative, so the limit stays on this branch. No disk about a curve point lies entirely on the curve.

(e) The region $xy \geq 1$, $x > 0$ is **closed**, not open. For a finite convergent sequence its limit has $x \geq 0$ and $xy \geq 1$, so $x = 0$ is impossible and the limit remains in the set. Boundary points with $xy = 1$ prevent openness.

Exercise 8.4.4: Five curved planar sets

(a) The circle $x^2 + y^2 = 1$ is **closed**, not open, by continuity of $x^2 + y^2$ and the absence of any disk contained in a circle.

(b) The region $x^2 + y^2 > 1$ is **open**, not closed: continuity preserves its strict inequality locally, but $(1 + 1/k, 0)$ tends to the excluded circle.

(c) The region $x^2 + y^2 \leq 1$, $y > 0$ is **neither**. The included arc point $(0, 1)$ is not interior; $(0, 1/k)$ tends to the excluded origin on the diameter.

(d) The region $y \geq x^2$ is **closed**, not open. It is defined by the nonnegative value of the continuous function $y - x^2$, and equality points have nearby points below the boundary.

(e) The region $y < x^2$ is **open**, not closed. It is a strict continuous inequality; for example $(0, -1/k)$ tends to the excluded origin.

Exercise 8.4.5: Compactness and convexity for the sets of Exercise 3

(a) The upper open half-plane is unbounded, hence not compact, but it is convex because averages of positive y-coordinates remain positive.

(b) The closed vertical segment is closed and bounded, hence compact. A segment is convex.

(c) The open vertical segment is not closed, hence not compact. It is convex: averaging two coordinates strictly between 1 and 3 keeps that coordinate strictly between the same bounds.

(d) The positive reciprocal curve is unbounded, hence not compact. It is not convex: the midpoint of $(1, 1)$ and $(2, 1/2)$ is $(3/2, 3/4)$, whose coordinate product is $9/8 \neq 1$.

(e) The region on and above the reciprocal curve is unbounded, hence not compact. It is convex: for positive x_1, x_2 and $0 \leq t \leq 1$,

$$\frac{1}{(1-t)x_1 + tx_2} \leq \frac{1-t}{x_1} + \frac{t}{x_2}.$$

This reciprocal convexity inequality, together with $y_i \geq 1/x_i$, puts each segment point on or above the same curve. Its horizontal coordinate stays positive. The inequality also follows by expanding the nonnegative quantity $t(1-t)(x_1 - x_2)^2$ after clearing positive denominators.

Exercise 8.4.6: Compactness and convexity for the sets of Exercise 4

(a) The circle is compact (closed and bounded) but not convex: $(1, 0)$ and $(-1, 0)$ have midpoint zero, off the circle.

(b) The exterior of the disk is unbounded and not compact. It is not convex: $(2, 0), (-2, 0)$ are in it but their midpoint is not.

(c) The upper half-disk with the diameter excluded is not closed, hence not compact. It is convex as the intersection of the closed unit disk and the convex open half-plane $y > 0$.

(d) The epigraph $y \geq x^2$ is unbounded, hence not compact. It is convex because $((1-t)x_1 + tx_2)^2 \leq (1-t)x_1^2 + tx_2^2 \leq (1-t)y_1 + ty_2$. The first inequality follows by subtracting the left side to get $t(1-t)(x_1 - x_2)^2 \geq 0$.

(e) The region $y < x^2$ is unbounded and not compact. It is not convex: $(-1, 1/2), (1, 1/2)$ are in it, but their midpoint $(0, 1/2)$ is not.

Exercise 8.4.7: Find the normal from difference vectors

Subtract the first point from every other listed point to get directions parallel to the hyperplane. Solve $D^T n = 0$, where these directions form the columns of D. One nonzero solution is

$$n = (0, 2, 3)^T, \qquad n^T p_1 = 11.$$

Checking gives $n^T(p_j - p_1) = 0$ for every other listed point. Thus (a) this n is a normal and (b) the required functional is $f(x) = n^T x$, with $d = 11$. The equation $n^T x = 11$ contains all the points. The difference vectors have rank 2, so their affine hull has the correct hyperplane dimension; the equation describes that whole hyperplane.

Exercise 8.4.8: Find the normal from difference vectors

Subtract the first point from every other listed point to get directions parallel to the hyperplane. Solve $D^T n = 0$, where these directions form the columns of D. One nonzero solution is

$$n = (-4, -3, 6)^T, \qquad n^T p_1 = 8.$$

Checking gives $n^T(p_j - p_1) = 0$ for every other listed point. Thus (a) this n is a normal and (b) the required functional is $f(x) = n^T x$, with $d = 8$. The equation $n^T x = 8$ contains all the points. The difference vectors have rank 2, so their affine hull has the correct hyperplane dimension; the equation describes that whole hyperplane.

Exercise 8.4.9: Find the normal from difference vectors

Subtract the first point from every other listed point to get directions parallel to the hyperplane. Solve $D^T n = 0$, where these directions form the columns of D. One nonzero solution is

$$n = (3, -1, 2, 1)^T, \qquad n^T p_1 = 5.$$

Checking gives $n^T(p_j - p_1) = 0$ for every other listed point. Thus (a) this n is a normal and (b) the required functional is $f(x) = n^T x$, with $d = 5$. The equation $n^T x = 5$ contains all the points. The difference vectors have rank 3, so their affine hull has the correct hyperplane dimension; the equation describes that whole hyperplane.

Exercise 8.4.10: Find the normal from difference vectors

Subtract the first point from every other listed point to get directions parallel to the hyperplane. Solve $D^T n = 0$, where these directions form the columns of D. One nonzero solution is

$$n = (-2, 3, -5, 1)^T, \qquad n^T p_1 = 4.$$

Checking gives $n^T(p_j - p_1) = 0$ for every other listed point. Thus (a) this n is a normal and (b) the required functional is $f(x) = n^T x$, with $d = 4$. The equation $n^T x = 4$ contains all the points. The difference vectors have rank 3, so their affine hull has the correct hyperplane dimension; the equation describes that whole hyperplane.

Exercise 8.4.11: Use the sign of the functional relative to its level

Here $f(x) = 2x_1 + x_2 + 5x_3 - x_4$ and $d = f(p) = 2 - 3 + 5 - 2 = 2$. At the origin $f(0) - d = -2 < 0$. For the three proposed points the functional values are $f(v_1) = 5$, $f(v_2) = -2$, and $f(v_3) = 2$. Thus v_2 is on the same open side as the origin, v_1 is on the opposite side, and v_3 lies in the hyperplane itself, on neither open side.

Exercise 8.4.12: Weak separation without strict parallel separation

With the prescribed normal $n = (3, 1, -2)$, use $f(x) = 3x_1 + x_2 - 2x_3$. Evaluation at the three points of A gives $-5, 4, 3$; at those of B it gives $7, 4, 6$. Hence $H = [f : 4]$ separates the sets weakly: $f(A) \le 4 \le f(B)$. But $\max f(A) = 4 = \min f(B)$, so there is no real number strictly between these two bounds. No hyperplane parallel to H strictly separates them. This only answers the prescribed-normal question; it does not assert that no differently oriented separating hyperplane could exist.

Exercise 8.4.13: The intersection of two hyperplanes in four dimensions

The two constants are $n_1 \cdot p_1 = 4$ and $n_2 \cdot p_2 = 22$. Thus solve

$$x_1 + 2x_2 + 4x_3 + 2x_4 = 4, \qquad 2x_1 + 3x_2 + x_3 + 5x_4 = 22.$$

Elimination gives $x_1 - 10x_3 + 4x_4 = 32$ and $x_2 + 7x_3 - x_4 = -14$. Take $x_3 = s, x_4 = t$ freely. The explicit answer is

$$x = (32, -14, 0, 0) + s(10, -7, 1, 0) + t(-4, 1, 0, 1), \qquad s, t \in \mathbb{R}.$$

Substitution checks both equations. The two direction vectors are independent because their third and fourth coordinates isolate s, t. Every solution was obtained through the two free variables, so this is the entire intersection, a two-dimensional flat.

Exercise 8.4.14: Possible dimensions of an intersection of flats

Choose $p \in F_1 \cap F_2$ and translate it to zero. Then $F_1 = p + U$, $F_2 = p + V$ with $\dim U = \dim V = 4$, and $F_1 \cap F_2 = p + (U \cap V)$. The dimension formula gives

$$\dim(U \cap V) = \dim U + \dim V - \dim(U + V) \ge 8 - 6 = 2,$$

and the intersection has dimension at most four. Each possibility occurs: take $U = \text{Span}\{e_1, e_2, e_3, e_4\}$ in $\mathbb{R}^6$, and take V respectively as $\text{Span}\{e_1, e_2, e_5, e_6\}$, $\text{Span}\{e_1, e_2, e_3, e_5\}$, or U. The intersection dimensions are $2, 3, 4$.

Exercise 8.4.15: An already implicit hyperplane equation

The equation $Ax = b$ is $x_1 - 3x_2 + 4x_3 - 2x_4 = 5$. Thus $f(x) = x_1 - 3x_2 + 4x_3 - 2x_4$ and $d = 5$. The coefficient vector is nonzero, so its level set is a hyperplane in $\mathbb{R}^4$.

Exercise 8.4.16: A null space of codimension one

Set $f(x) = 2x_1 + 5x_2 - 3x_3 + 6x_5$ and $d = 0$. Then $[f : 0]$ is exactly the given null space in $\mathbb{R}^5$. The fourth coordinate is free because its coefficient is zero; the functional itself is nonzero, so this is a hyperplane rather than the whole space.

Exercise 8.4.17: Describe a row plane by its perpendicular direction

Solve $Bn = 0$ for $B = \left(\begin{smallmatrix} 1 & 3 & 5 \\ 0 & 2 & 4 \end{smallmatrix}\right)$. The second equation gives $n_2 = -2n_3$, the first then gives $n_1 = n_3$. Take $n = (1, -2, 1)$. Both independent rows are perpendicular to n, so their two-dimensional span equals $\{x : n^T x = 0\}$. Thus $f(x) = x_1 - 2x_2 + x_3$, $d = 0$.

Exercise 8.4.18: A second row plane

For $B = \left(\begin{smallmatrix} 1 & 4 & -5 \\ 0 & -2 & 8 \end{smallmatrix}\right)$, solving $Bn = 0$ gives $n_2 = 4n_3$ and $n_1 = -11n_3$. Choose $n = (-11, 4, 1)$. Since the two rows of B are independent, their span is the plane $-11x_1 + 4x_2 + x_3 = 0$. Take $f(x) = -11x_1 + 4x_2 + x_3$ and $d = 0$.

Exercise 8.4.19: A column plane and a transpose null space

A normal must be perpendicular to both columns, so solve $B^T n = 0$: $n_1 + 4n_2 - 7n_3 = 0$, $2n_2 - 6n_3 = 0$. These give $n_2 = 3n_3$, $n_1 = -5n_3$. Thus $n = (-5, 3, 1)$ is a normal. The columns are independent, so the column space is the full plane $-5x_1 + 3x_2 + x_3 = 0$. Take this expression as f and take $d = 0$.

Exercise 8.4.20: A second column-plane equation

The normal equations are $n_1 + 5n_2 - 4n_3 = 0$ and $2n_2 - 4n_3 = 0$. Choose $n_3 = 1$, giving $n = (-6, 2, 1)$. The two original columns are independent, so $H = \text{Col}\, B$ equals $\{x : -6x_1 + 2x_2 + x_3 = 0\}$. Hence $f(x) = -6x_1 + 2x_2 + x_3$, $d = 0$.

Exercise 8.4.21: Four assertions about functionals and separation

(a) **False.** A linear functional maps $\mathbb{R}^n$ *to* $\mathbb{R}$, not generally from $\mathbb{R}$ to $\mathbb{R}^n$.

(b) **False.** The correct general formula is $f(x) = n^T x$, a dot product with a fixed vector, not scalar multiplication kx, which has vector output when $n > 1$. For example $f(x_1, x_2) = x_1 + x_2$ is a functional.

(c) **True.** A point in both sets would have its functional value simultaneously strictly below and strictly above the same level d. That is impossible.

(d) **False.** Two disjoint closed convex sets need not have a positive separating gap if neither is compact. For example $A = \{(x, y) : y \geq e^x\}$ and $B = \{(x, 0) : x \in \mathbb{R}\}$ are closed, convex, and disjoint. Any functional bounded on one side along the entire line B must have zero x-coefficient. Separation would then have to be horizontal, but A has positive y values arbitrarily close to zero, while B has $y = 0$. No horizontal level lies strictly between the sets.

Exercise 8.4.22: Four further assertions about separating hyperplanes

(a) **True.** A nonzero functional has a nonzero coefficient vector, so each of its real level sets is a hyperplane.

(b) **False.** The normal must be nonzero. If $n = 0, d = 0$, the set is all of $\mathbb{R}^n$; if $n = 0, d \neq 0$, it is empty. Neither is a hyperplane.

(c) **False.** Convexity is missing. Take $A = \{0\}$ and B the unit circle. They are disjoint, with A compact and B closed, but the center cannot be strictly separated from all points of the surrounding circle. If $b \in B$, both b and $-b$ belong to it, so no functional can put both strictly on the same side of a level separating them from zero.

(d) **False.** Failure of one chosen hyperplane does not imply failure of every choice. Take $A = \{(0, 0)\}$, $B = \{(2, 0)\}$, and the line $y = 100$, which does not strictly separate them. Their convex hulls are still disjoint, and the line $x = 1$ does strictly separate them.

Exercise 8.4.23: Separate a point from a triangle by an edge normal

Take $f(x, y) = 3x - 2y$. Its values at the three listed triangle vertices are $1, 9, 9$, whereas $f(p) = 10$ for $p = (4, 1)$. Every convex combination of the vertices therefore has value at most 9. Thus the line $[f : 19/2]$, or $3x - 2y = 9.5$, strictly separates the triangle from p: all its hull values are below 9.5, and the point's value is above.

Exercise 8.4.24: A second point outside a triangle

Take $f(x, y) = -2x + 3y$. At $(1, 2), (5, 1), (4, 4)$ the values are $4, -7, 4$; at $p = (2, 3)$ the value is 5. Thus $[f : 9/2]$ strictly separates p from the triangle, since its entire convex hull has functional value at most $4 < 4.5 < 5$.

Exercise 8.4.25: Separate two open balls using their center direction

Let $p = (4, 1)$, $f(x) = p \cdot x = 4x_1 + x_2$, and $v = (3/4)p = (3, 3/4)$. Set $d = f(v) = 51/4$. For $x \in B(0, 3)$, Cauchy–Schwarz gives $f(x) < 3\sqrt{17} < 51/4$. For $x \in B(p, 1)$, it gives $f(x) = 17 + p \cdot (x - p) > 17 - \sqrt{17} > 51/4$. Thus $[f : 51/4]$ strictly separates the balls. The hinted point v lies in neither

ball because $\|v\| = 3\sqrt{17}/4 > 3$ and $\|v - p\| = \sqrt{17}/4 > 1$. The strict inequalities are justified by $\sqrt{17} < 17/4$ and $3\sqrt{17} < 51/4$, which follow on squaring positive quantities.

Exercise 8.4.26: Translate the two-ball construction

The center difference is $n = p - q = (4, -2)$, with $\|n\| = \sqrt{20}$. Take $f(x) = 4x_1 - 2x_2$, so $f(q) = 2$, $f(p) = 22$. The point $v = q + (3/4)(p - q) = (5, 3/2)$ has $f(v) = 17$. For $x \in B(q, 3)$, $f(x) < 2 + 3\sqrt{20} < 17$; for $x \in B(p, 1)$, $f(x) > 22 - \sqrt{20} > 17$. Therefore $[f : 17]$ strictly separates the balls. The inequalities follow from $\sqrt{20} < 5$, and they leave a positive gap between the sets.

Exercise 8.4.27: A closed set whose convex hull loses closedness

Let

$$S = \{(n, 1/n), (-n, 1/n) : n = 1, 2, 3, \ldots\}.$$

This set is closed: any convergent sequence in it is bounded, so its integer first coordinates range through only finitely many possibilities; its limit is in that finite subset of S. But the midpoint of $(n, 1/n)$ and $(-n, 1/n)$ is $(0, 1/n)$, so these points lie in $\operatorname{conv} S$ and tend to $(0, 0)$. No convex combination of points in S has height zero, because it is a finite weighted average of strictly positive heights. Thus $(0, 0)$ is not in the convex hull, which is consequently not closed.

Exercise 8.4.28: Disjoint hulls need not give a strict separating gap

Take the closed set $B = S$ of Exercise 8.4.27 and the compact set $A = \{(0, 0)\}$. Every point in $\operatorname{conv} B$ has positive height, so the two convex hulls are disjoint. Suppose a hyperplane strictly separated A and B. After changing the functional's sign if necessary, its level would satisfy $0 = f(0) < d < f(b)$ for every $b \in B$. Taking the midpoint of the two points with first coordinates $\pm n$ would give $f(0, 1/n) > d$. Letting $n \to \infty$ contradicts $d > 0$, since $f(0, 1/n) \to 0$. Thus strict separation is impossible. The compact-set separation theorem cannot be applied to $\operatorname{conv} B$ as a closed set here: Exercise 8.4.27 shows that it is not closed.

Exercise 8.4.29: An open ball is convex

For $x, y \in B(p, \delta)$ and $0 \le t \le 1$, the triangle inequality gives

$$\|(1 - t)x + ty - p\| = \|(1 - t)(x - p) + t(y - p)\| \le (1 - t)\|x - p\| + t\|y - p\| < \delta.$$

At $t = 0$ or 1 this is the original strict inequality; between them it is a weighted average of two numbers strictly below δ. Thus the whole segment remains inside the open ball.

Exercise 8.4.30: Convex hull preserves boundedness

Since S is bounded, choose R with $\|s\| \le R$ for every $s \in S$. For a convex combination $y = \sum_i \lambda_i s_i$, use the triangle inequality, nonnegativity, and sum-one property to obtain

$$\|y\| \le \sum_i \lambda_i \|s_i\| \le \sum_i \lambda_i R = R.$$

Hence the same ball bounds $\operatorname{conv} S$. If S is empty its hull is empty and bounded as well.